

Trees

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DA-IICT

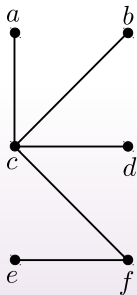
Chapter Summary

- 1 Trees introduction
- 2 Tree traversal

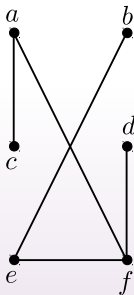
Trees

Tree: A tree is a connected simple undirected graph with no simple circuits.

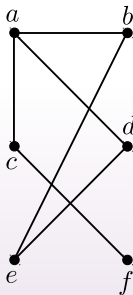
Examples:



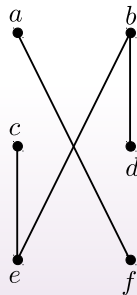
(a) Tree



(b) Tree



(c) Not a tree



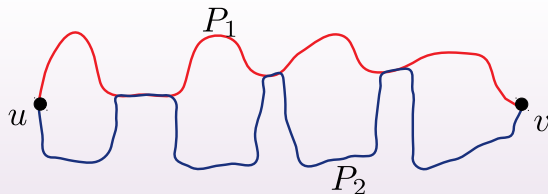
(d) Forest

Some important facts about trees

Theorem: An undirected graph G is a tree if and only if there is a unique simple path between each pair of vertices of G .

Proof:

- G is a tree.
- There is a unique path between each pair of vertices.
- Suppose not
- Let $u - - v$, there is no unique path. i.e., at least 2 paths.



- $u - - v$ there is a circuit.
- A contradiction.

Proof contd . . .

- Assume that there is a unique simple path between each pair of vertices of G .
- Clearly, G is connected.



- Need to prove G has no cycle.
- Assume that G has a circuit.
- Let $u - - - v$ there be a circuit.

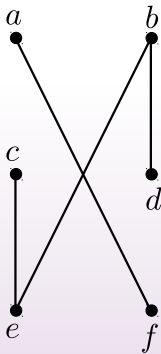


- There are at least 2 paths between u and v .

Forest

Forest: A forest is a (not necessarily connected) simple undirected graph with no simple circuits. (Collection of trees).

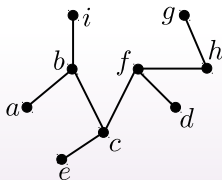
Example:



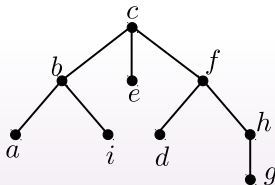
Rooted trees

Rooted tree: It is a tree in which there is a vertex designated as the root and every vertex is directed away from the root.

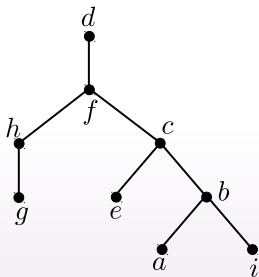
Example:



A tree



A rooted tree with root *c*



A rooted tree with root *d*

Note: For different vertex as the root, different rooted tree is generated.

Different tree terminologies

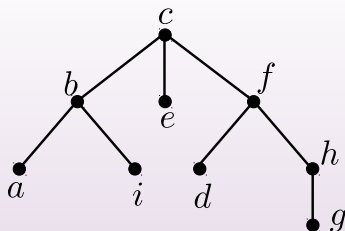
Let (T, r) be a rooted tree with root at r

Parent: The parent of a vertex a (other than the root) is a vertex w such that (w, a) is a directed edge in the rooted tree.

Child: a is called the child of w

Sibling: Children of the same parent.

Ancestor: parent, grand parent, grand grand parent, ...

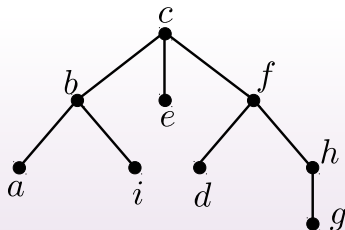


Different tree terminologies

Internal vertices: vertices that have at least one child.

Leaves: Vertices that have no children.

Descendants: The descendants of a vertex v are the vertices that have v as an ancestor.

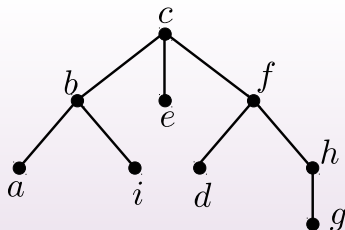


Different tree terminologies

Depth: The number of ancestors of a node v is called the depth of v .

Height of a node: The height of a node v is the maximum depth from the node v to any leaf.

Height of a tree: The height of a tree is the maximum depth from the root to any leaf.

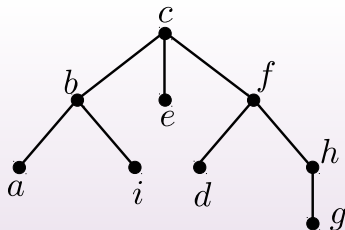


Different tree terminologies

Subtree: Subtree of a tree T is the tree rooted at some node in T .

The subtree rooted at v is the subgraph containing v and all its descendants, and all the directed edges between them.

Level: The length of the unique path from the root to a vertex is called the level of that vertex.



Some important results

Theorem: Every tree with n vertices has exactly $n - 1$ edges.

Proof: We prove it by induction.

Base step:

- Let $n = 1$, i.e., there is only one vertex.
- There is no edge. So, $n - n = 0$ edges.
- A tree with 1 vertex has 0 edges.

Inductive step:

- Assume that for some value $k = n$ the statement is true.
- A tree with n vertices has $n - 1$ edges.
- We have to prove that $n + 1$ vertices have $n + 1 - 1 = n$ edges.
- Consider a tree T' by removing a leaf vertex in T
- T' is a tree with n vertices.
- By inductive hypothesis, T' has $n - 1$ edges.
- So T has $n + 1$ vertices and $n - 1 + 1 = n$ edges.

Degree

Degree of a vertex: No of children of that vertex.

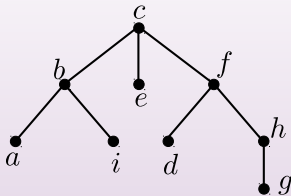
Degree of a tree: maximum degree of any vertex in the tree.

m -ary tree: In a rooted tree where every internal vertex has at most m children.

$m = 2$: binary tree.

Full m -ary tree: In a rooted tree where every internal vertex has exactly m children.

Balanced tree: All leaves are at the level h or $h - 1$, where h is the height of the tree.



A few results

Theorem: A full m -ary tree with i internal vertices has $mi + 1$ vertices.

Proof: Every vertex other than the root is a child of an internal vertex. Thus, there are $m \cdot i$ such children, plus 1 root.

Theorem: A full m -ary tree with

- 1 n vertices has $i = \frac{n-1}{m}$ internal vertices and $\ell = \frac{(m-1)n+1}{m}$ leaves.
- 2 i internal vertices has $n = mi + 1$ vertices and $\ell = (m-1)i + 1$ leaves.
- 3 ℓ leaves has $n = \frac{m\ell-1}{m-1}$ vertices and $i = \frac{\ell-1}{m-1}$ internal vertices.

Theorem: There are at most m^h leaves in an m -ary tree of height h .

Proof: By induction on $h \geq 0$.

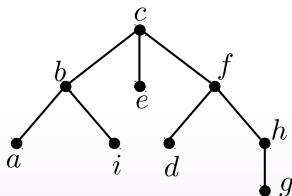
Corollary: If an m -ary tree of height h has ℓ leaves then $h \geq \lceil \log_m \ell \rceil$.

Proof: Since $\ell \leq m^h$, we have $\log_m \ell \leq h$. But h is a non-negative integer, so $\lceil \log_m \ell \rceil \leq h$.

Ordered tree

Ordered tree: A rooted tree in which the set of children of each vertex is assigned a total order.

In a drawing of an ordered rooted tree, with the root shown at the top, the children of a vertex are shown from left to right.



In an ordered binary tree (usually called just a binary tree), if an internal vertex has two children, the first child is called the **left child** and the second child is called the **right child**.

left subtree: The tree rooted at the left child of a vertex.

Right subtree: The tree rooted at the right child of a vertex..

Tree traversal

- Information is stored in ordered rooted tree
- Methods are needed for visiting each vertex of the tree in order to access data

Tree traversal: A procedure used to systematically visit each vertex in a tree is called a traversal algorithm, or a traversal.

- Three most commonly used traversal algorithms are:
 - 1 Preorder Traversal
 - 2 Inorder Traversal
 - 3 Postorder Traversal
- Each of these algorithms can be defined recursively.

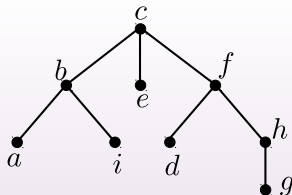
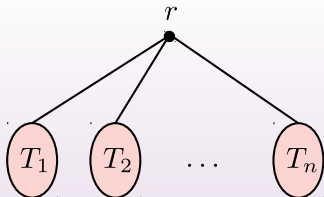
Preorder traversal

Let T be an ordered rooted tree with root r .

If T consists only of r , then r is the preorder traversal of T .

Otherwise, suppose that $T_1, T_2, \dots, T_n$ are the subtrees at r from left to right in T . The preorder traversal begins by visiting r . It continues by traversing T_1 in preorder, then T_2 in preorder, and so on, until T_n is traversed in preorder.

Example:



Preorder traversal:

$c, \quad b, a, i, \quad e, \quad f, d, h, g$

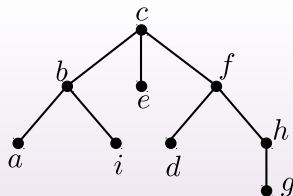
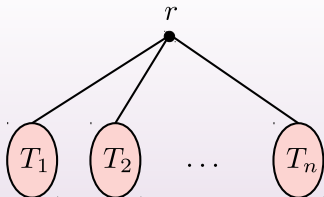
Inorder traversal

Let T be an ordered rooted tree with root r .

If T consists only of r , then r is the inorder traversal of T .

Otherwise, suppose that $T_1, T_2, \dots, T_n$ are the subtrees at r from left to right in T . The inorder traversal begins by traversing T_1 in inorder, then visiting r . It continues by traversing T_2 in inorder, then T_3 in inorder, $\dots$, and finally T_n in inorder.

Example:



Inorder traversal:

$a, b, i, \quad c, \quad e, \quad d, f, g, h$

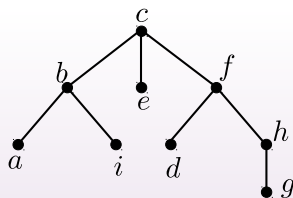
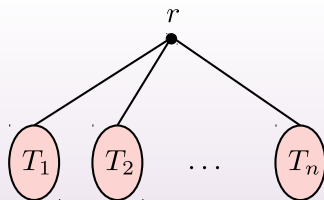
Postorder traversal

Let T be an ordered rooted tree with root r .

If T consists only of r , then r is the postorder traversal of T .

Otherwise, suppose that $T_1, T_2, \dots, T_n$ are the subtrees at r from left to right in T . The postorder traversal begins by traversing T_1 in postorder, then T_2 in postorder, $\dots$, then T_n in postorder, and ends by visiting r .

Example:



Postorder traversal:

$a, i, b, \quad e, \quad d, g, h, f, \quad c$